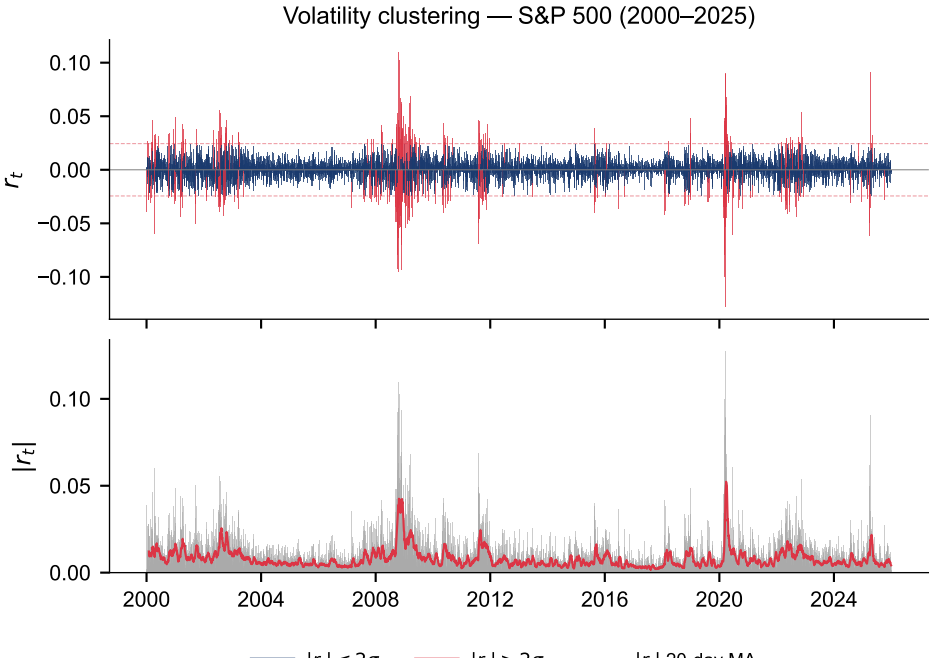
Volatility clustering — S&P 500 (2000–2025)
r_t
0.10
0.05
0.00
−0.05
−0.10
$|r_t|$
0.10
0.05
0.00
2000
2004
2008
2012
2016
2020
2024
$|r_t| \leq 2\sigma$
$|r_t| > 2\sigma$
$|r_t|$ 20-day MA